

MONTGOMERY, AL, USA

WMO: 722260

Lat: 32.300N

Lon: 86.408W

Elev: 202

StdP: 14.59

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13895

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.2	27.9	9.6	9.0	32.4	14.2	11.4	35.1	22.6	49.3	20.2	50.3	5.3	310	0.385

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	96.7	76.1	94.7	76.0	92.9	75.7	79.6	90.8	78.5	89.3	77.8	88.3	6.7	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.7	140.0	84.1	75.7	135.6	83.3	74.9	132.0	82.5	43.1	90.9	42.1	89.7	41.2	88.4	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.6	16.5	14.2	DB	17.6	99.0	3.8	2.7	14.8	101.0	12.6	102.6	10.4	104.1	7.6	106.1	
			WB	15.9	81.6	3.7	2.1	13.3	83.1	11.1	84.4	9.0	85.6	6.3	87.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.1	47.6	52.2	58.8	65.6	73.4	79.8	82.2	82.1	77.4	67.0	56.4	50.2	(6)
	DBStd	14.27	10.21	9.58	8.66	6.64	5.49	3.67	2.77	3.19	5.08	7.74	8.48	9.42	(7)
	HDD50	428	170	81	25	1	0	0	0	0	0	1	32	118	(8)
	HDD65	2039	542	369	226	73	9	0	0	0	2	72	279	467	(9)
	CDD50	6322	97	143	299	470	724	895	999	995	823	529	225	123	(10)
	CDD65	2457	4	11	34	92	268	445	534	530	375	135	22	7	(11)
	CDH74	23985	17	57	306	874	2558	4477	5529	5410	3407	1159	158	33	(12)
	CDH80	10618	0	4	45	204	990	2132	2698	2659	1514	350	21	1	(13)
(14) Wind	WSAvg	5.8	7.0	7.2	7.1	6.4	5.3	4.8	4.6	4.5	5.3	5.4	5.7	6.5	(14)
(15) Precipitation	PrecAvg	52.1	4.6	5.2	5.8	4.2	3.9	4.0	5.1	3.7	4.0	2.6	3.9	5.1	(15)
	PrecMax	76.5	11.2	13.4	12.4	15.6	12.8	14.4	10.0	10.4	9.4	7.6	10.2	14.1	(16)
	PrecMin	35.5	1.1	1.1	0.9	0.5	0.2	0.3	1.3	0.7	0.6	0.0	0.5	1.0	(17)
	PrecStd	9.8	2.2	2.6	2.9	3.0	2.6	2.7	2.2	1.9	2.2	1.9	2.2	2.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.4	79.9	84.6	87.3	93.1	97.6	98.5	100.1	96.9	90.3	82.9	78.1	(19)
		MCWB	64.5	66.6	66.3	69.0	73.4	75.3	76.5	75.2	74.4	71.9	68.5	66.9	(20)
	2%	DB	71.9	75.3	80.7	85.0	90.5	95.2	96.0	96.7	93.9	87.0	78.3	73.7	(21)
		MCWB	63.0	64.3	65.0	68.3	72.5	75.2	77.0	76.4	73.9	70.3	66.0	65.4	(22)
	5%	DB	68.2	71.6	77.5	82.4	88.2	93.1	93.9	94.3	91.3	84.0	75.0	69.9	(23)
		MCWB	61.2	61.6	63.4	67.3	71.7	75.0	76.9	76.5	73.9	69.2	64.2	63.4	(24)
	10%	DB	64.3	68.2	74.0	79.5	85.9	90.6	91.8	91.9	88.5	80.8	71.7	65.9	(25)
		MCWB	58.6	60.4	61.7	65.5	70.8	74.6	76.5	76.3	73.3	67.7	62.8	60.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.8	69.5	70.1	73.2	77.1	79.9	81.1	81.2	78.6	76.3	72.8	70.3	(27)
		MCDB	72.3	76.5	77.4	82.8	87.1	90.6	92.3	92.5	88.5	84.0	77.9	74.8	(28)
	2%	WB	65.3	66.9	68.1	71.3	75.3	78.4	79.7	79.5	77.3	74.5	69.5	67.1	(29)
		MCDB	70.1	72.5	75.8	80.3	85.2	89.4	91.1	90.7	87.3	82.0	74.2	71.4	(30)
	5%	WB	62.6	64.7	66.2	69.8	74.1	77.4	78.6	78.4	76.3	72.7	67.3	64.5	(31)
		MCDB	67.0	70.2	74.3	77.9	83.7	88.1	89.5	89.1	85.7	79.3	73.1	68.7	(32)
	10%	WB	59.7	62.1	64.1	68.2	72.9	76.4	77.8	77.6	75.3	70.7	64.3	61.6	(33)
		MCDB	63.9	66.8	71.6	76.1	82.2	86.4	88.4	87.9	84.1	76.9	69.6	65.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	21.2	21.7	22.8	23.5	22.2	20.1	19.3	19.7	20.5	23.1	23.9	20.8	(35)
		MCDBR	23.8	24.2	25.9	25.6	24.1	23.6	22.0	22.7	23.4	24.9	25.7	23.5	(36)
	5% WB	MCWBR	15.8	14.4	12.7	11.4	9.1	7.6	7.1	7.0	7.7	10.8	14.6	15.9	(37)
		MCWBR	19.8	20.3	21.0	20.3	20.5	19.9	19.3	19.8	19.1	18.3	20.3	19.7	(38)
(40) Clear-Sky Solar Irradiance	taub		0.324	0.331	0.360	0.390	0.440	0.478	0.536	0.525	0.433	0.374	0.347	0.326	(40)
		taud	2.521	2.504	2.428	2.336	2.244	2.187	2.032	2.059	2.301	2.448	2.464	2.526	(41)
	Ebn at Noon		285	294	292	285	271	259	244	245	265	275	273	278	(42)
		Edh at Noon	27	30	35	40	44	47	55	52	39	32	29	26	(43)
(44) All-Sky Solar Radiation	RadAvg		889	1070	1438	1795	1974	1954	1897	1761	1563	1330	1005	763	(44)
	RadStd		62	130	130	130	139	147	129	102	131	163	95	83	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (40 neighbors)	+0.99	N/A	N/A	N/A	N/A	N/A	-83	-182	+353	+247
		+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+275	+173

Nomenclature: See separate page